

Database Design Document

WEEK 2: ENTITY , ATTRIBUTES AND ENTITY RELATIONSHIP DIAGRAM

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Project Title

Customized Phone Case Ordering System

❖ Entity Identification

1. Customer

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- First_Name
- Last_Name
- Email
- Phone_Number
- Password
- Address

2. Product (Phone Case)

Primary Key: Product_ID

Attributes:

- Product_ID (PK)
- Product_Name
- Brand
- Phone_Model
- Material

- Price
- Stock_Quantity

3. Customization

Primary Key: Customization_ID

Attributes:

- Customization_ID (PK)
- Uploaded_Image
- Custom_Text
- Font_Style
- Font_Color
- Background_Color
- Preview_Image

4. Cart

Primary Key: Cart_ID

Attributes:

- Cart_ID (PK)
- Customer_ID (FK)
- Total_Amount

5. Order

Primary Key: Order_ID

Attributes:

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Order_Status

- Total_Amount

6. Payment

Primary Key: Payment_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Status
- Payment_Date
- Amount

7. Delivery

Primary Key: Delivery_ID

Attributes:

- Delivery_ID (PK)
- Order_ID (FK)
- Delivery_Address
- Courier_Name
- Tracking_Number
- Delivery_Status
- Delivery_Date

8. Admin

Primary Key: Admin_ID

Attributes:

- Admin_ID (PK)
- Admin_Name
- Email
- Password
- Role
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❖ Relationship Table

Entity 1	Relationship	Entity 2	Cardinality
Customer	places	Order	One-to-Many (1:M)
Customer	owns	Cart	One-to-One (1:1)
Cart	contains	Product	Many-to-Many (M:N)
Product	has	Customization	One-to-Many (1:M)
Customer	creates	Customization	One-to-Many (1:M)
Order	has	Payment	One-to-One (1:1)
Order	has	Delivery	One-to-One (1:1)
Admin	manages	Product	One-to-Many (1:M)
Admin	manages	Order	One-to-Many (1:M)

Foreign Keys

- **Customer_ID → Cart**
- **Customer_ID → Order**

- Order_ID → Payment
- Order_ID → Delivery

Final Entities

1. Customer
2. Product
3. Customization
4. Cart
5. Order
6. Payment
7. Delivery
8. Admin

❖ **Entity-Relationship Diagram:**

ER DIAGRAM - CUSTOMIZED PHONE CASE ORDERING SYSTEM

